AgentGS: Empowering Earth Science Agents with Plan Composition and Completion over Skill Hypergraphs

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Abstract

Complex geoscientific tasks often require multiple tools to work together and exchange intermediate results across computational steps. Existing methods commonly plan around specific tools, tying workflows to particular implementations, whereas reusing entire strategies makes it difficult to combine local steps from different strategies according to task requirements. We present AgentGS, a method for capability composition and tool orchestration over skill hypergraphs. AgentGS uses a three-layer Skill–Capability–Tool representation to decouple task strategies, reusable operations, and concrete tools, allowing it to organize an operational workflow before selecting its implementations. The skill hypergraph connects strategies through shared capabilities and provides a structured basis for composition across strategies. Given selected strategies, the system expands their required operations and prerequisite steps. When the current operation chain lacks necessary intermediate steps, it searches for feasible connecting branches along dependencies declared by other strategies. By jointly checking step dependencies, data compatibility, and tool availability, the system incorporates branches that satisfy the constraints without ambiguity, thereby reusing local steps from other strategies. It then selects concrete tools, propagates task data and intermediate results, and locally recovers failed steps. AgentGS can also organize newly composed capability workflows into candidate Skills for subsequent registration and reuse. On GeoToolEval, which contains 603 evaluation records across ten geoscientific domains, AgentGS achieves a mean Final Answer Score (FAS) of 0.8731. Configuration comparisons show higher scores for the full method on the reported overall and multistep process metrics. Some configurations provide only partial controls or did not actually activate the intended ablation, so their differences cannot be directly attributed to the corresponding modules.

CCS Concepts

• **Computing methodologies** → **Intelligent agents**; *Planning and scheduling*.

Keywords

LLM Agents, Compositional Planning, Tool Orchestration, Skill Hypergraph, Scientific Agent

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1 Introduction

Earth science seeks to derive testable scientific conclusions from observational data, from climate change and critical minerals to energy resources and disaster risk. Its research frontiers are data intensive, computationally demanding, and interdisciplinary, with increasing emphasis on predictive systems science driven by big data and artificial intelligence [15]. Progress requires organizing distributed data, tools, and methods into reproducible, traceable analytical workflows.

Such workflows typically require data retrieval, spatial or temporal alignment, variable derivation, statistical modeling, and result verification. Researchers manually connect these stages, despite differences in data formats and reporting standards. Scientific agents built around large language models (LLMs) offer a way to automate this process by translating natural-language goals into multistep plans and coordinating tools [4, 5]. Crucially, later geoscientific operations depend on artifacts produced by earlier steps. Reliable tool orchestration is therefore essential to completing the analysis.

Existing geoscientific agents support multistep execution using frameworks such as ReAct [12]. GeoAgent [1] and Earth-Agent [5], for example, use planning–execution–reflection architectures. However, their execution units are typically individual calls, individual tools, or entire Skills. Tool-level planning couples workflows to concrete interfaces, whereas whole-Skill reuse makes it difficult to extract and recombine local operations from different strategies. An unresolved problem is how to compose operations across strategies while preserving their prerequisites and usage constraints, and subsequently select suitable tool implementations.

Evaluation presents a related challenge. EarthSE assesses geoscientific reasoning through questions and exploratory dialogue without requiring actual tool execution [11]. GeoBenchX [6], EarthBench [5], and GeoAgentBench [13] examine multistep tool use; the latter two also assess processes and outcomes. We evaluate tool composition and orchestration across geoscientific themes, distinguishing reference-answer agreement from completion of required operations within the same task. Failure analysis additionally requires parameters, artifacts, and invocation records; a matching answer or successful individual call does not establish a correct analysis.

We introduce AgentGS and GeoToolEval to address these challenges. Our contributions are:

- AgentGS separates Skills, Capabilities, and Tools and completes missing capabilities through a hypergraph. This supports operation reuse across strategies, constraint preservation, and executable tool selection.

- GeoToolEval comprises 603 evaluation records across ten geoscientific research-frontier themes. Single-step and multistep tasks support joint analysis of tool dependencies, composition, intermediate-artifact use, and final answers.
- AgentGS achieves a mean Final Answer Score (FAS) of 0.8731 on GeoToolEval, outperforming the evaluated models and agents. Multistep strata and configuration comparisons show answer and workflow differences, whose causes require execution traces and scrutiny of actual interventions.

2 Related Work

Planning and dependency representations. ReAct interleaves reasoning, actions, and observations [12]. ToolPlanner methods support task execution over large tool collections [8, 10], while LLM+P and Generalised BDI Planning introduce explicit planning and goal management [7, 9]. GeoAgent and Earth-Agent apply multistep planning and execution to geoscientific analysis [1, 5]. SciToolAgent orchestrates scientific tools through a knowledge graph [4]; HyperAgent constructs a tool hypergraph from input/output schemas and searches for supporting tools [14]. AgentGS instead represents shared operations as Capabilities and strategies as Skills. It preserves each operation's dependencies and usage constraints within its source strategy, allowing local cross-strategy reuse before tool selection.

Geoscientific evaluation. GeoBench and EarthSE evaluate knowledge understanding and scientific exploration without requiring actual tool calls [2, 11]. Scientific agents demonstrate execution of code and data-processing tasks [1, 4, 5]. More recent benchmarks explicitly examine orchestration: GeoBenchX evaluates multistage geospatial tasks against reference solutions [6]; Earth-Bench distinguishes autonomous planning from instruction following and records tool inputs and outputs [5]; GeoAgentBench checks invocation order, parameters, and map artifacts in a GIS sandbox [13]; and GeoNatureAgent Benchmark tests multitool analysis and error recovery with real APIs [3]. GeoToolEval extends evaluation across ten research themes and supports analysis of operation selection, dependencies, composition, artifact consumption, and final answers within the same task. It also examines completion of necessary operations, helping distinguish missing-capability failures from execution errors.

3 Overview of AgentGS

3.1 Overall Workflow

Given a natural-language goal q and resources X, AgentGS retrieves Skills by task description, domain, and input types. SkillSet stores strategies; GeoSkillKG connects them through shared Capabilities. Applicable Skills guide the Planner to select operations, expand mandatory predecessors, and complete missing operations or connecting branches. When no Skill applies, GeoToolKG supplies candidates for direct tool-call planning. Both routes share execution and verification (Figure 1(a)).

After capability planning, Tool Grounding selects implementations through Capability–Tool mappings, aided by tool associations and an LLM within the candidate scope. The Executor binds resources and successful upstream artifacts. The Judge checks whether results satisfy the goal, returning an answer or failure feedback for recovery of unfinished steps. Verified new compositions can be registered as candidate Skills in SkillSet and GeoSkillKG for subsequent retrieval.

3.2 Hierarchical Representation and Skill Hypergraphs

A *Skill* is a reusable task strategy with an objective, applicable inputs, expected artifacts, capability steps, constraints, validation requirements, and recovery policies. Each step specifies a Capability, mandatory predecessors, optional or mandatory status, and permitted tools. Task-relevant local steps can be selected without importing the entire strategy. New Skills package composed workflows without creating new Capability definitions or Tool implementations.

A *Capability* defines a shared operation: its function, input and output types, preconditions, postconditions, and candidate tools. Sharing does not remove strategy-specific constraints; deep-time and carbon-storage Skills may share raster analysis while permitting different implementations. A *Tool* is a callable interface with an identifier, parameter schema, input/output definitions, and runtime information. Concrete resources and artifacts are bound during execution. Skill–Capability membership and Capability–Tool mappings are many-to-many; mappings additionally record operation and parameter constraints.

The task-specific plan uses capability instances:

$$P = (V, D_d, D_c, \beta), \qquad v_i = (c_i, s_i, r_i). \tag{1}$$

Here, c_i, s_i, and r_i identify the operation, source strategy, and step/input role. D_d and D_c record data and control dependencies, and β binds resources or upstream artifacts. Operations on different objects remain distinct instances even when they use the same tool.

The skill hypergraph connects registered Skills through shared Capabilities:

$$H = (C, E), \qquad E = \{e_s : s \in S\}, \qquad e_s \subseteq C. \tag{2}$$

Capabilities are nodes and each Skill's capability set is a hyperedge. Skills containing $\{a, b\}$ and $\{b, c\}$ share b. Membership alone does not determine execution order; dependency conditions must be expanded. A provenance-bearing relation $d = (s, U, c)$ specifies that c requires all predecessors in U within Skill s.

3.3 Strategy Retrieval and Initial Planning

AgentGS retrieves Skills using the task description, domain, and input types, selects applicable strategies, and chooses capabilities covering task requirements. An LLM assists selection within the retrieved candidates. Hypergraph Composition expands mandatory predecessors and constructs P, retaining provenance, candidate-tool scopes, and input references.

3.4 Missing-Capability Completion and Bridging

Missing Capability Completion checks for missing required operations, unsatisfied inputs, and unavailable implementations. Missing operations are supported through activated Skills or directly compatible Skills. Unsatisfied input dependencies may trigger multihop

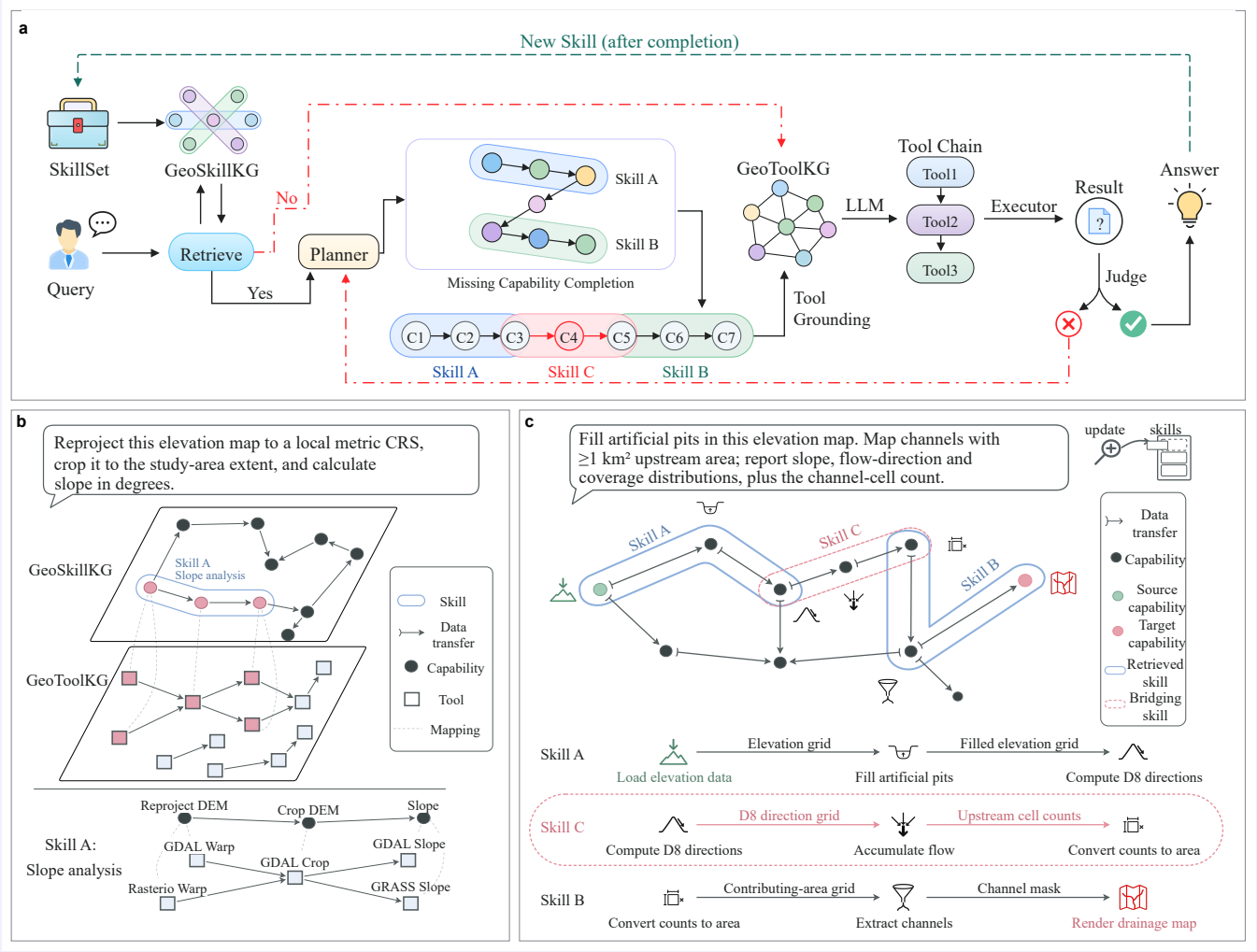


Figure 1: AgentGS hierarchical representation and overall workflow. (a) Retrieval, planning, completion, execution, verification, and Skill writeback. (b) Shared capabilities, strategy-specific dependencies, and tool implementations. (c) A connecting branch retrieved from another Skill. Verified compositions can be saved as candidate Skills.

search along explicit compatibility relations. After insertion, the system rechecks prerequisites and dependency conflicts. Directed bridging provides a separate route through inactive Skills.

3.4.1 Directed Bridging with a Fixed Active Boundary. Suppose activated Skills A and B supply disconnected fragments. Let u be an output-boundary capability of A, with no successor among its active dependencies, and v an input-boundary capability of B, with no predecessor among its active dependencies. If inactive Skill C declares $u \rightarrow m \rightarrow v$, this branch can supply intermediate capability m without importing the entire Skill.

The active capability set is fixed at search start. Each branch must come from a single inactive Skill and contain at least one inactive intermediate capability. Membership first identifies Skills containing both endpoints; explicit dependencies or applicable conditional-ordering relations then determine directed simple paths. Co-occurrence and list order do not establish direction. An existing indirect data path from u to v excludes the candidate pair; a direct connection alone does not. Newly inserted capabilities do not recursively extend the current search.

Candidate branches must satisfy five checks:

(1) **Direction:** each adjacent pair follows an explicit dependency or applicable conditional order.
(2) **Prerequisites:** all required predecessors belong to the fixed active set or candidate branch.
(3) **Types:** upstream outputs and downstream inputs share at least one concrete type, excluding generic labels such as `Any` and `Parameter`.
(4) **Implementations:** branch operations and required predecessors have available tools within permitted scopes. The

Algorithm 1 Directed Bridging from a Single Provider

Require: Plan P after compatibility completion; registered knowledge; available tools
Ensure: Updated P and acceptance/rejection records
1: Return if bridging is disabled, recovery is active, or P has already been searched
2: Fix active capabilities; mark P as searched
3: Enumerate boundary pairs from different Skills; skip pairs with an indirect data path
4: Search inactive Skills for single-provider branches satisfying the five validity checks
5: Deduplicate by provider/path; reject ambiguous pairs; check joint acyclicity
6: Attempt insertion, preserving entry constraints and updating exit dependencies, tools, and bindings
7: Recompose; reject external additions or unmet mandatory dependencies

entry and external active predecessors retain their original scopes; other branch nodes use the provider Skill's constraints.

(5) **Acyclicity:** both individual and joint insertions preserve acyclic dependencies.

For each ordered endpoint pair, only a unique valid branch may be inserted automatically. Multiple candidates produce an ambiguity report and halt completion for that pair. Insertion preserves the entry instance's dependencies and tools, updates the exit's dependencies, tool scope, and bindings, and recomposes the plan. A branch is rejected if recomposition adds capabilities outside the branch or leaves mandatory dependencies unsatisfied. Algorithm 1 summarizes the procedure.

3.5 Tool Grounding, Execution, and Recovery

GeoToolKG records tool functions, interfaces, provenance, and recommended transitions. ToolGraph maps Capabilities to Tools in both directions and records operation and parameter constraints. The completed plan filters implementations by availability and instance scope; parameter schemas and semantic input/output types guide evaluation.

Tool Grounding filters implementations by capability requirements, instance provenance, and retry constraints, using an LLM within the candidate scope when needed. Tool Execution preferentially binds task resources, values, and upstream artifacts through rules. Artifacts must come from successful preceding steps and have matching types. Binding failure blocks invocation. The final answer is generated from structured execution results.

Failure Recovery handles the failed capability instance locally while retaining successful preceding results. Defaults permit at most three attempts per tool and one local substitution. A substitute must implement the same capability, lie within the instance's permitted scope, and pass fresh parameter/artifact checks. Calls with uncertain outcomes, partial execution, or schema-invalid outputs are not automatically replayed. Recovery remains within the existing capability plan; execution is suspended if continuing requires another capability.

3.6 Result Verification and Skill Writeback

The system separately records tool execution, answer generation, and task completion. Verification compares task requirements with successful results and artifacts. Unmet requirements or failures return to the Planner under recovery rules; incomplete answers identify obtained results and unfinished operations.

Verified new workflows can be packaged as candidate Skills recording objectives, applicable inputs, artifacts, steps, dependencies, permitted tools, and constraints. Distinct input objects remain distinct steps, with source provenance and usage constraints retained. After consistency checks, candidates are registered in SkillSet and GeoSkillKG. Reuse still requires applicability checks, step selection, resource binding, and tool choice.

4 GeoToolEval

GeoToolEval associates answer evaluation and execution-process analysis within each task. For example, statistics must consume the composite raster rather than the original images; successful calls alone do not establish the requested analysis. Its records contain a natural-language question, input resources, a reference answer and answer type, and reference tool calls. Single-step tasks assess data location, parameter determination, and basic invocation; multistep tasks examine operation composition, dependencies, and use of intermediate results. Tasks span ten research themes, with theme sizes determined by available resources and screening outcomes.

4.1 Sources and Construction

Tasks are either selected from existing question-answer resources or derived from scientific data. Geoscientific big-data analysis includes 40 selected GeoQuestions1089 tasks involving physical geographic objects and spatial operations. Remote sensing includes 49 of Earth-Bench's 248 tasks, retaining source questions, choices, and answers while adapting calls to our interfaces. Critical-mineral and climate-change tasks are selected from habitable_earth_dataset; underlying resources include MRDS, paleoclimate reconstructions, GISTEMP, and SSP warming summaries. Data-derived tasks specify identifiable objects, variables, and computational relationships, with answers determined through processing and calculation. Table 1 lists the data sources.

Reference calls map source operational requirements to registered tools and record parameters and input/output information. Existing tasks retain source identifiers or resource locators; derived tasks retain data-location fields, derivation records, and computational scripts. Quality control checks traceability, record completeness, consistency of reference invocations, and sample execution. Sample validation is recorded separately from full-benchmark evaluation and does not establish correctness on all tasks.

4.2 Composition and Evaluation

The benchmark contains 603 evaluation records: 424 have one reference call, 119 have two or three, and 60 have at least four. Its 179 multistep records constitute 29.68% of the benchmark. Nine cross-theme questions occur in both critical minerals and climate change, so 603 does not denote 603 distinct questions.

Table 1: GeoToolEval task sources and reference tool-invocation chain lengths. Counts refer to evaluation records.

Task theme	Main data or question sources	1 step	2–3	≥ 4	Total
Three-dimensional Earth structure	LITHO1.0, SEMUCB-WM1, USGS earthquake catalog	85	15	0	100
Geoscientific big-data analysis	Selected GeoQuestions1089 subset	40	0	0	40
Critical minerals	Habitable-Earth questions; MRDS and cross-theme climate data	44	14	0	58
Climate change	Habitable-Earth questions; paleoclimate, GISTEMP, SSP, and cross-theme mineral data	31	16	1	48
Energy resources	MRDS, GEM, OPSD, World Bank, and population data	100	0	0	100
Global ocean carbon cycle	SOCAT, GLODAP, SNAPO-CO_2	100	0	0	100
Deep-time biosphere	BioDeepTime	12	7	5	24
Carbon capture and storage	Synthetic SCCS-OFPOSDU CCS–MRV data	12	5	7	24
Remote sensing	Selected Earth-Bench subset	0	7	42	49
Natural-disaster event statistics	EM-DAT International Disaster Database	0	55	5	60
Total		**424**	**119**	**60**	**603**

A step denotes one reference tool invocation, rather than one scientific operation or a direct measure of scientific difficulty. Energy-resource and ocean-carbon task wrappers, for example, can perform several operations internally. Chain-length strata therefore describe orchestration demands under the current interfaces. A reference chain represents one implementation under the current interfaces. Domain and chain-length groups describe task backgrounds and interface-level workflow composition, respectively.

Final Answer Score (FAS).. FAS lies in $[0, 1]$ and retains partial credit. Scoring normalizes choices, matches dates and identifiers, applies numerical tolerances, scores matches among multiple values/text items, and compares structured fields. Empty answers receive zero. For N records,

$$\overline{FAS} = \frac{1}{N}\sum_{i=1}^{N} FAS_i. \tag{3}$$

The main experiment uses $N = 603$. Groups average their own records with failures retained. This is a task-level micro-average, distinct from a domain-balanced macro-average or full-credit rate.

Tool Selection F1 (TSF1). Let G_i and A_i be the reference and actual successful-call sequences for task i. Treating them as multisets, match calls one to one by tool name, retaining multiplicity. With m_i matches,

$$\mathrm{TSF1}_i = \frac{2m_i}{|G_i| + |A_i|}. \tag{4}$$

This is the harmonic mean of precision $m_i/|A_i|$ and recall $m_i/|G_i|$. Missing, irrelevant, and repeated successful calls affect the score.

Tool Order Score (TOS).. Using the longest common subsequence,

$$\mathrm{TOS}_i = \frac{\mathrm{LCS}(G_i, A_i)}{|G_i|}. \tag{5}$$

For read–filter–count versus read–count–filter, TSF1 is 1 and TOS is 2/3. Extra calls do not directly reduce TOS, so TSF1 is also needed to assess excess invocations. Both scores are zero when no call succeeds. Failed calls are excluded from successful sequences, but failed tasks remain in the averages. All three metrics use per-task averaging and are reported separately. Subset ablations do not summarize all 603 records.

TSF1 and TOS assess tool coverage and reference order. Matching names and order can still conceal wrong parameters or artifact consumption. Execution traces and computational results are therefore required; a different valid workflow is not a failure solely because its process metrics are lower.

5 Experiments

We compare answers and tool-workflow consistency while distinguishing configuration differences from attributable module effects. The illustrative case explains the mechanism and does not substitute for generalization or runtime-cost experiments.

5.1 Setup and Main Results

The main experiment covers all 603 GeoToolEval records; configuration comparisons use a fixed 57-record subset. FAS, TSF1, and TOS average per-task scores with failures retained. AgentGS uses GPT-5.6 Terra; main FAS values come from its first complete run, retaining partial credit. GeoAgent and EarthAgent use the same backbone. Direct-answer baselines are DeepSeek V4 Pro, GPT-5.6 Terra, GPT-5.4, Qwen-Plus, and GeoGPT. Process metrics compare successful calls from the three agents in the same tool environment against the same reference chains; direct-answer models have no tool traces.

AgentGS obtains FAS of 0.8731 versus GeoAgent's 0.7804 and direct GPT-5.6 Terra's 0.3137 (Table 2). Its present end-to-end run over 603 records yields TSF1/TOS of 0.9170/0.9166, including execution failures and incomplete final answers. The table reports 0.8663/0.7911 for GeoAgent and 0.1182/0.1390 for EarthAgent. These scores measure reference-tool coverage and relative order; correct inputs, artifacts, and scientific results require additional checks.

Single-step, two- to three-step, and at-least-four-step FAS values are 0.9425, 0.7557, and 0.6167. On 119 two- to three-step records, TSF1/TOS are 0.9779/0.9832; on 60 longer records, they are 0.8013/0.7785. Lower long-chain process scores indicate remaining coverage and ordering difficulties. Longer chains are concentrated in remote sensing, so domain and chain-length effects overlap (Figure 2).

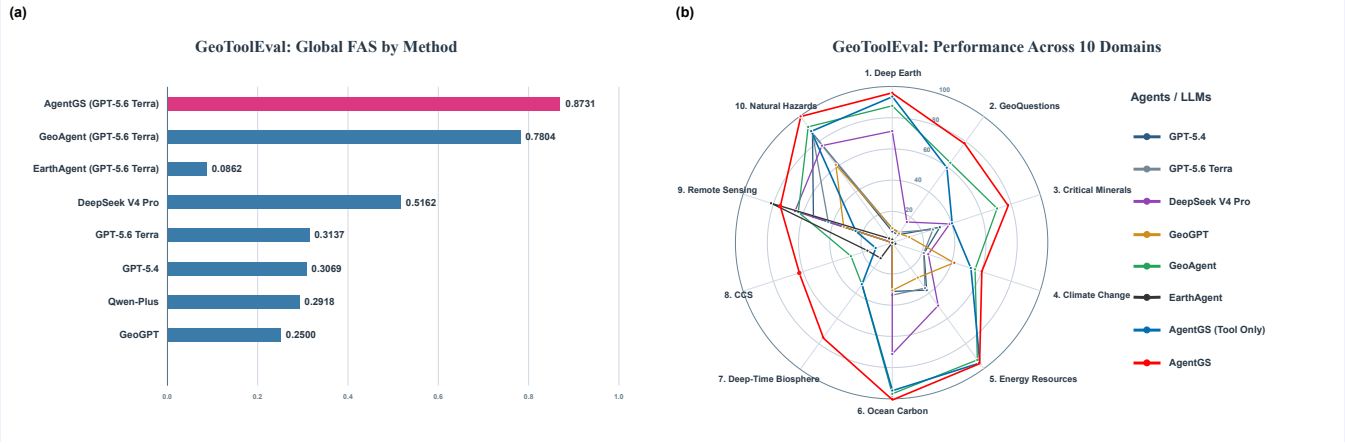


Figure 2: Overall and domain-specific performance of AgentGS and baselines on GeoToolEval: (a) mean FAS across all evaluation records; (b) mean FAS across the ten geoscientific domains.

Table 2: GeoToolEval baseline comparison. AgentGS process metrics are per-record averages from the present 603-record end-to-end experiment; FAS values are retained from the source manuscript. Direct-answer language models have no tool traces; their process metrics are not applicable (—).

Method	FAS	TSF1	TOS
AgentGS (GPT-5.6 Terra)	**0.8731**	0.9170	0.9166
GeoAgent (GPT-5.6 Terra)	0.7804	0.8663	0.7911
EarthAgent (GPT-5.6 Terra)	0.0862	0.1182	0.1390
DeepSeek V4 Pro	0.5162	—	—
GPT-5.6 Terra	0.3137	—	—
GPT-5.4	0.3069	—	—
Qwen-Plus	0.2918	—	—
GeoGPT	0.2500	—	—

Table 3: Evaluation results on GeoBenchX. Both methods use GPT-5.6 Terra.

Method	Mean score (0–2) ↑	Full-match rate (%) ↑
ReAct	1.1980	53.47
AgentGS	**1.3762**	**62.87**

5.2 External Evaluation on GeoBenchX

We integrate official GeoBenchX tools [6] and construct dedicated Skills, Capabilities, dependencies, and mappings. AgentGS and the official ReAct baseline are each reported on 202 tasks using GPT-5.6 Terra. That model is also the single judge: 0, 1, and 2 denote mismatch, partial match, and match to reference solutions. Mean score and full-match rate are separate from GeoToolEval FAS.

AgentGS scores 1.3762 versus 1.1980 for ReAct, a difference of 0.1782 points; full-match rates are 62.87% versus 53.47%, a difference of 9.40 percentage points. Although task counts are supplied, exact task membership and consistent experimental conditions still require verification before interpreting these differences as transfer evidence.

5.3 Ablation Configuration Comparisons

The subset covers all ten domains, with 22 single-step, 20 two- to three-step, and 15 longer records; multistep tasks comprise 61.40%. The design uses Full on the same subset as reference, but actual intervention scopes are limited: A1 is a partial control, A2's intended ablation was not activated, and A5 changes only part of ranking. Differences therefore do not uniformly represent controlled single-module effects.

A1 is designed to plan at the tool level with the same strategies, dependencies, and tools. A2 is designed to disable mandatory-predecessor expansion while retaining selection and dependency declarations; completion may compensate for omissions. A3 retains gap diagnosis but disables supporting-capability insertion, including bridging. A4 disables only bridging. A5 is designed to choose the first valid tool in a fixed order while retaining requirements, scopes, and validation. A6 disables local recovery, retaining planning, grounding, and binding before the first attempt.

A1's FAS decreases from Full are 16.39/12.38 percentage points; A2's are 16.27/12.38 and A3's are 17.63/15.31 (Table 4). Multistep decreases are smaller in all three cases. The partial A1 control and inactive A2 ablation prevent direct attribution to hierarchical planning or predecessor expansion. The comparison does not show greater multistep effects for predecessor expansion or independent completion.

A4's overall/multistep decreases are 2.27/4.20 points; A6's are 4.15/7.20. Attribution to bridging or recovery requires inserted branches, failure/recovery traces, and consistent conditions. A5 increases overall FAS from 0.6387 to 0.6415 but lowers multistep FAS from 0.5333 to 0.5184, giving different directions for its partial ranking change.

Full TSF1/TOS are 0.9249/0.9392 overall and 0.8862/0.8952 on 35 multistep records (Table 5). Full uses the supplementary experiment; A1–A6 use formal worksheet records. All six have lower overall and

Table 4: Configuration comparisons on the fixed 57-record subset. A1 is a partial control, A2 did not activate the intended ablation, and A5 changes only part of tool ranking. ΔFAS is the full method's mean FAS minus the corresponding configuration's mean FAS, in percentage points; positive values indicate lower performance than the full method.

Configuration	Mean FAS	Multistep FAS	ΔFAS (pp)
Full method	0.6387	0.5333	—
A1: Direct tool-level planning	0.4748	0.4095	16.39
A2: No predecessor expansion	0.4760	0.4095	16.27
A3: No independent completion	0.4624	0.3802	17.63
A4: No directed bridging	0.6160	0.4913	2.27
A5: No tool-selection optimization	0.6415	0.5184	−0.28
A6: No execution recovery	0.5972	0.4613	4.15

Table 5: Ablation process metrics. Multistep scores cover the 35 subset records with at least two reference calls. Each column averages per-task scores over its corresponding records. Full uses the supplementary 57-task experiment; A1–A6 use per-task records from the formal 57-task worksheet.

Configuration	Mean TSF1	Mean TOS	Multistep TSF1	Multistep TOS
Full method	0.9249	0.9392	0.8862	0.8952
A1: Direct tool-level planning	0.8599	0.8778	0.8147	0.8295
A2: No predecessor expansion	0.8694	0.8958	0.8302	0.8588
A3: No independent completion	0.8732	0.8865	0.8364	0.8438
A4: No directed bridging	0.8603	0.8997	0.8249	0.8652
A5: No tool-selection optimization	0.8503	0.8851	0.8133	0.8414
A6: No execution recovery	0.8208	0.8588	0.7652	0.7985

multistep process scores. Multistep changes from Full are −7.15/−6.57 points for A1, −5.60/−3.64 for A2, and −4.98/−5.14 for A3. A4 changes by −6.13/−3.00 and has lower TSF1 but higher TOS than A3. A5 changes by −7.29/−5.38, reflecting partial ranking adjustment. A6 changes by −12.10/−9.67, with a larger coverage decrease. These are configuration differences; parameters, artifact consumption, and module effects require traces and scrutiny of data provenance.

5.4 Illustrative Case Study

Consider several rasters of the same region and variable, with aligned grids and a reference value in the same units. The task requires a pixel-wise mean composite, its valid-pixel mean, the difference from the reference, and the comparison relation.

Skill Retrieval returns RasterCompose (A) and ReferenceCompare (B). Planning selects InputCheck (w) and RasterMean (u) from A, and ScalarDiff (v) and SignCheck (z) from B, yielding $w \to u$ and $v \to z$. Composition expands required predecessors while retaining provenance and tool scopes. The fragments remain disconnected: u produces a raster file, whereas v requires a numerical value.

Completion retrieves $u \to m \to v$ from inactive RasterDiff (C), where ValidMean (m) computes the composite's valid-pixel mean. The query explicitly requires this statistic. Under the example's assumptions, this is the unique branch satisfying prerequisites, types, availability, and acyclicity. Insertion retains u's constraints, adds m, updates v's dependencies, tools, and bindings, and preserves $w \to u$ and $v \to z$, forming $w \to u \to m \to v \to z$ (Figure 3).

Grounding selects permitted validation, compositing, statistics, differencing, and comparison tools. Bridge grounding requires unique endpoint instances with the correct order. Execution passes the validated raster list to compositing, then passes a list containing only the composite path to statistics. The returned mean and task reference bind to a and b for $d = a - b$; sign comparison evaluates d against zero. Thus, each output concerns the same composite and reference.

If statistics fails, local recovery retains validation and compositing and reuses the artifact. Uncertain calls are not replayed, and recovery requiring another capability suspends execution. Successful results yield the raster, mean, difference, and relation. After verification, the workflow can form candidate CompositeCompare (D), with rebindable image-list/reference inputs, artifact dependencies, and usage constraints. Its applicability and reuse benefit on other tasks require further validation.

6 Discussion and Conclusion

AgentGS separates strategies, operations, and implementations to compose local capabilities across Skills. Hypergraph completion uses dependencies with provenance and type/tool constraints; grounding, artifact binding, and local recovery yield executable calls. Verified compositions can form candidate Skills for reuse. AgentGS's main FAS exceeds the evaluated baselines, and configuration comparisons show answer and process-score differences. A1's partial control, A2's inactive ablation, A5's partial ranking intervention, and metrics drawn from different materials prevent

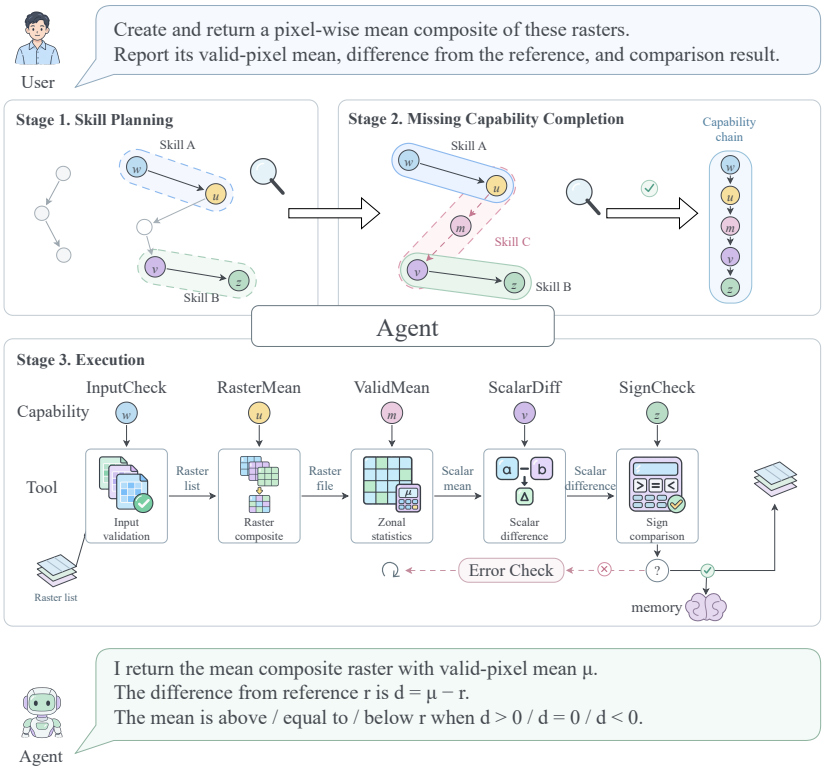


Figure 3: Illustrative cross-Skill completion. Selected fragments $w \to u$ and $v \to z$ are connected through inactive Skill C's $u \to m \to v$ branch. The resulting five-step workflow is grounded in tools with explicit artifact propagation.

interpreting all differences as independent causal module contributions. Scientific correctness additionally requires execution traces.

Coverage depends on registered knowledge. Structural and type checks do not establish task necessity, and recovery is limited to the existing plan. Candidate Skills do not expand underlying capability/tool coverage; their effects on later performance and planning cost remain untested. Future work will integrate task semantics and feedback into completion, assess Skill reuse across tasks, and test unseen combinations and tool changes.

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